① Simplex Tableau

- You are able to solve LPs using the simplex tableau.

Understand each part of the simplex tableau: → 不要忘记负号

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

- The upper left corner is the reduced cost vector.
- The upper right corner is the negative of the objective value.
- The bottom part are the transformed constraints and the corresponding basic variables.
- One can find $\boldsymbol{A}_B^{-1} \boldsymbol{A}$ in the tableau. We can also often get $\boldsymbol{A}_B^{-1}$.

② Two-Phase Method

The goal of the two-phase method is to find an initial BFS for the original problem.

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbb{1}$ means an all-one vector, $\boldsymbol{b} \geq 0$).

$$\begin{aligned} \underset{\boldsymbol{x},\boldsymbol{y}}{\text{minimize}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq 0. \end{aligned}$$

Observation:

- Solving this auxiliary problem will give a BFS for the original problem (or tell that the original problem is infeasible if the optimal value of the auxiliary problem is not 0).

Two-Phase Method in Simplex Tableau

We can use the two-phase method, when there is no obvious initial basic feasible solution.

Phase I:

- We need to form the initial tableau for Phase I.
- We need to compute the top row (reduced costs):
  - For basic part, the reduced costs are 0.
  - For nonbasic part, $\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j = -\mathbb{1}^\top \boldsymbol{A}_j$, so the jth reduced cost is the negative of the sum of that column.
- This also applies to the initial objective value, which equals the negative of the sum of the right hand side vector.

Two-Phase Method in Simplex Tableau

Suppose we have finished the Phase I simplex tableau operations, resulting in an optimal solution $(\hat{\boldsymbol{x}}, 0)$ with optimal value 0.

To carry out Phase II in the simplex tableau:

1. Drop all the columns associated with the auxiliary variables.
2. Change the top row to the reduced costs corresponding to the original problem by using the reduced cost formula:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j$$

   and compute the current (negative) objective value.
3. Continue the simplex tableau to solve the original problem.

If the optimal solution of the auxiliary problem is degenerate, then replace the basic indices of included auxiliary variables with indices of original variables to form a complete basis.

可能遇到 degeneracy 的问题

例：

$$\begin{aligned} \text{minimize} \quad & x_1 + 3x_2 + x_4 - 2x_5 \\ \text{subject to} \quad & x_1 + 2x_2 + 4x_4 + x_5 = 2 \\ & x_1 + 2x_2 - 2x_4 + x_5 = 2 \\ & -x_1 - 4x_2 + 3x_3 = 1 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned} \Rightarrow$$

$$\begin{aligned} \text{minimize} \quad & y_1 + y_2 + y_3 \\ \text{subject to} \quad & x_1 + 2x_2 + 4x_4 + x_5 + y_1 = 2 \\ & x_1 + 2x_2 - 2x_4 + x_5 + y_2 = 2 \\ & -x_1 - 4x_2 + 3x_3 + y_3 = 1 \\ & x_1, x_2, x_3, x_4, x_5, y_1, y_2, y_3 \geq 0, \end{aligned}$$

⇓

B	0	0	0	6	0	2	0	1	0
1	1	2	0	4	1	1	0	0	2
7	0	0	0	−6	0	−1	1	0	0
3	0	$-\frac{2}{3}$	1	$\frac{4}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{3}$	1

$$A = \begin{bmatrix} 1 & 2 & 0 & 4 & 1 \\ 1 & 2 & 0 & -2 & 1 \\ -1 & -4 & 3 & 0 & 0 \end{bmatrix}$$

Phase Ⅱ：

B	0	1	0	0	−3	−2
1	1	2	0	0	1	2
4	0	0	0	1	0	0
3	0	$-\frac{2}{3}$	1	0	$\frac{1}{3}$	1

从 2, 4, 5 列中找与 1, 3 两列 linear independent 的

Degeneracy & Bland's Rule

Degeneracy

We call a basic feasible solution $\boldsymbol{x}$ (non)degenerate if (none) some of the basic variables are 0.

- To ensure that there is no cycling, we use the smallest index rule when choosing the entering and exiting index.
- If the smallest index rule is used, one does not need to worry about the degenerate cases in the simplex method.

- The dual optimal solution comes as a by-product when we use the simplex algorithm. The term

$x = A_B^{-1} b$ and $y = (\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$

Weak Duality Theorem

Weak Duality Theorem (Primal $\rightsquigarrow$ min, Dual $\rightsquigarrow$ max)

If $\boldsymbol{x}$ is feasible for the primal problem and $\boldsymbol{y}$ is feasible for the dual, then we have:

$$\boldsymbol{b}^\top \boldsymbol{y} \leq \boldsymbol{c}^\top \boldsymbol{x}.$$

Consequences:

- A primal (dual) feasible point can be used to construct bounds for the objective value of the dual (primal) problem.
- If one is unbounded, then the other one must be infeasible. △
- Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problems, respectively. If $\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{b}^\top \boldsymbol{y}$, then $\boldsymbol{x}$ and $\boldsymbol{y}$ are optimal solutions of the primal and dual, respectively.

Strong Duality Theorem

$c^T x = b^T y$

Strong Duality Theorem

If a linear program has an optimal solution, so does its dual and the optimal values of the primal and dual problems are equal.

All possible states of a pair of primal/dual linear programs are:

P | D	Finite Optimum	Unbounded	Infeasible
Finite Optimum	✓		
Unbounded		✗	✓
Infeasible		✓	✓

- If both the primal and dual are feasible, then both problems must have optimal solutions. (This is one way to quickly determine if an LP is bounded).

Complementarity Conditions

Consider the primal-dual pair:

Primal			Dual		
min	$\boldsymbol{c}^\top \boldsymbol{x}$		max	$\boldsymbol{b}^\top \boldsymbol{y}$	
s.t.	$\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i,$	$i \in M_1,$	s.t.	$y_i \geq 0,$	$i \in M_1$
	$\boldsymbol{a}_i^\top \boldsymbol{x} \leq b_i,$	$i \in M_2,$		$y_i \leq 0,$	$i \in M_2$
	$\boldsymbol{a}_i^\top \boldsymbol{x} = b_i,$	$i \in M_3,$		y_i free,	$i \in M_3$
	$x_j \geq 0,$	$j \in N_1,$		$\boldsymbol{A}_j^\top \boldsymbol{y} \leq c_j,$	$j \in N_1$
	$x_j \leq 0,$	$j \in N_2,$		$\boldsymbol{A}_j^\top \boldsymbol{y} \geq c_j,$	$j \in N_2$
	x_j free,	$j \in N_3,$		$\boldsymbol{A}_j^\top \boldsymbol{y} = c_j,$	$j \in N_3$

Theorem: Complementarity Conditions

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problem. Then $\boldsymbol{x}$ and $\boldsymbol{y}$ are optimal if and only if:

$$y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x} - b_i) = 0, \quad \forall\, i \quad \text{and} \quad x_j \cdot (\boldsymbol{A}_j^\top \boldsymbol{y} - c_j) = 0, \quad \forall\, j.$$

Sensitivity analysis

Case 1: change b

$$\tilde{b} = b + \Delta b$$

$$\tilde{x}_B = A_B^{-1}(b + \Delta b) = x^* + \lambda A_B^{-1} e_j \geq 0$$

Case 2: change c

① $j \in N$

$$c_N' = c_N^T + \lambda e_j^T - c_B^T A_B^{-1} A_N = r_N^T + \lambda e_j^T$$

② $j \in B$

$$c_N' = c_N^T - (c_B^T + \lambda e_j^T) A_B^{-1} A_N = r_N^T - \lambda e_j^T A_B^{-1} A_N$$

Case 3: change A

If the change is for a number in a non-basic column, say A_j, then the original optimal solution is still feasible, the only change is to the reduced cost of jth variable.

If the change is for a number in a basic column, then nearly all the numbers in the tableau will change. In general, there is not a simple way to deal with it.

找 A_B^{-1} 和 $A_B^{-1}A_N$

Consider the following linear program:

$$
\begin{array}{lrcrcrcrcl}
\text{minimize} & x_1 & + & x_2 & + & 2x_3 & + & x_4 & & \\
\text{subject to} & x_1 & + & 2x_2 & - & x_3 & + & x_4 & = & 2 \\
& 2x_1 & & & + & x_3 & - & x_4 & \leq & 2 \\
& -x_1 & & & - & 2x_3 & + & x_4 & \geq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0.
\end{array}
$$

Denote $\boldsymbol{x} = (x_1, x_2, x_3, x_4, s_1, s_2)$ as the decision variable to the standard form of the above problem, where s_1, s_2 are the slack variables corresponding to the second and third constraints. The following table gives the final simplex tableau when solving the standard form of the above problem:

B	1	0	$\frac{7}{2}$	0	0	$\frac{1}{2}$	$-\frac{3}{2}$
2	1	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	$\frac{1}{2}$
4	-1	0	-2	1	0	-1	1
5	1	0	-1	0	1	-1	3

Primal – Dual 求 Dual Problem

① $\max \quad c^T x$

$s.t. \quad 1^T x = k$

$x \geq 0$

$x \leq 1$

② $\min \quad c^T x$

$s.t. \quad Ax \leq b$

$Cx = d$

③ $\min \quad c^T x$

$s.t. \quad Ax - s = b$

$s \geq 0$

④ $\min \quad c^T x^+ - c^T x^-$

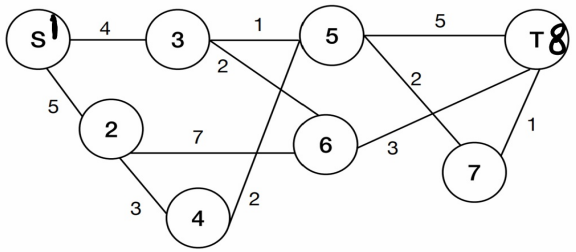
$s.t. \quad Ax^+ - Ax^- \geq b$

$x^+ \geq 0$

$x^- \geq 0$

几类模型：

① The shortest path problem



② Transportation problem

From/To	1	2	3	4	5
1	–	20	13	11	28
2	20	–	18	8	46
3	13	18	–	9	27
4	11	8	9	–	20
5	28	46	27	20	–

Table 1: Cost of moving a car

	1	2	3	4	5
Present	110	335	400	420	610
Need	150	200	600	200	390

Table 2: Number of current and needed cars

Let x_{ij} denote the number of cars moving from place i to j.

$$
\begin{array}{ll}
\text{minimize} & \sum_{i=1}^{5}\sum_{j=1}^{5} c_{ij}x_{ij} \\
\text{s.t.} & \sum_{i\neq 1} x_{i1} - \sum_{j\neq 1} x_{1j} \geq 150 - 110 \\
& \sum_{i\neq 2} x_{i2} - \sum_{j\neq 2} x_{2j} \geq 200 - 335 \\
& \sum_{i\neq 3} x_{i3} - \sum_{j\neq 3} x_{3j} \geq 600 - 400 \\
& \sum_{i\neq 4} x_{i4} - \sum_{j\neq 4} x_{4j} \geq 200 - 420 \\
& \sum_{i\neq 5} x_{i5} - \sum_{j\neq 5} x_{5j} \geq 390 - 610 \\
& x_{ij} \geq 0 \quad \forall i, j \\
& x_{ij} \in \mathbb{Z} \quad \forall i, j.
\end{array}
$$

供需关系

③ 见习题

一些证明思想

① For polyhedron, a feasible point is a basic solution if and only if it is an extreme point.

联想 convex

如果不是extreme point，则可用 $\lambda x + (1-\lambda)y$ 表示

② 常用替换（转置）

$x^TA^T = (Ax)^T$

$(Ax)^Ty = y^TAx$

$c^Tx = b^Ty \Rightarrow x^TA^Ty = c^Tx$

$y^Tb = y^TAx = b^Ty$

③ 很多证明会用到 weak / strong duality 来推矛盾或进行不等式变换